

Why Do CF and Review Models Disagree? User-Specific Rating–Review Scale Gaps

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Abstract

Ratings and reviews capture different aspects of preference, so rating-based collaborative-filtering and review-based models may disagree on a user–item interaction. Aggregate accuracy does not reveal why they disagree or which user characteristics are associated with it. We examine whether this reflects discrepancies between users’ rating tendencies and review tone. We introduce the rating–review scale gap (RRS gap), the standardized difference between a user’s rating leniency and review-tone bias. Large disagreement alone does not identify the better model, whereas the RRS gap is consistently associated with whether the review-based model predicts above or below the rating-based model, with domain-dependent strength. Such user-specific discrepancies may therefore help explain model disagreement. Our source code is available at https://github.com/alsrb9936/rating_and_reviews.

CCS Concepts

• Information systems → Recommender systems.

Keywords

review-based recommendation, rating–review discrepancy, prediction disagreement

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1 Introduction

Review text enriches recommendation beyond scalar ratings [3, 6, 12]. Ratings summarize overall preference within a limited numerical scale, whereas reviews selectively describe particular aspects, experiences, and attitudes. For example, two users may assign the same rating to an item, yet one may describe it as offering good value for money, emphasizing a positive aspect, while the other may focus only on how inconvenient it was to use, emphasizing a negative aspect. Conversely, users may express similarly positive or negative experiences in their reviews but assign different ratings because of differences in rating standards or leniency. Thus, the same rating need not correspond to the same review tone, and similar review tone need not correspond to the same rating [1, 7, 10]. Rating-history and review-history models may accordingly assign different scores to the same user–item pair. We study this disagreement as an interaction-level model-selection region because aggregate error can obscure where review-based prediction helps or hurts. Building on work on rating–sentiment discrepancy [1, 4, 11], we examine whether persistent differences between user-specific **rating leniency** and rating-conditioned **review tone** explain disagreement between separately trained NeuMF [5] and ReviewOnly models. We define their standardized difference as the **rating–review scale gap (RRS gap)** and test whether it tracks the direction and benefit of model disagreement.

2 Disagreement Measures

ReviewOnly constructs user and item review representations from training reviews and predicts ratings from these representations. Each training review t is encoded as $\phi(t) \in \mathbb{R}^d$ with SBERT [9]. Separate projection networks transform the mean user and item representations, and a prediction network outputs the rating:

$$\begin{aligned} \mathbf{v}_u^{\text{rev}} &= |\mathcal{R}_u^{\text{tr}}|^{-1} \sum_{t \in \mathcal{R}_u^{\text{tr}}} \phi(t), & \mathbf{v}_i^{\text{rev}} &= |\mathcal{R}_i^{\text{tr}}|^{-1} \sum_{t \in \mathcal{R}_i^{\text{tr}}} \phi(t), \\ \hat{r}_{ui}^{\text{rev}} &= f_{\theta}([g_u(\mathbf{v}_u^{\text{rev}}) \| g_i(\mathbf{v}_i^{\text{rev}})]). \end{aligned} \quad (1)$$

Here, $g_u(\cdot)$ and $g_i(\cdot)$ denote the user and item projection layers, respectively. Validation and test target reviews are excluded during prediction, and ratings supervise the prediction network.

For each test interaction, we define $G_{ui} = \hat{r}_{ui}^{\text{rev}} - \hat{r}_{ui}^{\text{cf}}$ and $M_{ui} = |G_{ui}|$, and partition interactions into quartiles of M_{ui} separately within each domain. Let $e_{ui}^{\text{cf}} = |r_{ui} - \hat{r}_{ui}^{\text{cf}}|$ and $e_{ui}^{\text{rev}} = |r_{ui} - \hat{r}_{ui}^{\text{rev}}|$. Positive transfer (PT) occurs when $e_{ui}^{\text{rev}} < e_{ui}^{\text{cf}} - 0.05$, negative transfer (NT) occurs when $e_{ui}^{\text{rev}} > e_{ui}^{\text{cf}} + 0.05$, and the remaining interactions are Neutral. Within each analysis group, ΔMAE is ReviewOnly MAE minus NeuMF MAE.

3 User-Specific Rating–Review Scale Gap

Let r_{ui} be user u 's rating for item i . We assign each review an expected sentiment score $s_{ui} \in [1, 5]$ using a frozen RoBERTa-based five-class sentiment model [2]. All user-bias estimates use the training split. Let $\mathcal{D}_u^{\text{tr}}$ contain n_u interactions. For review tone, $\mu_r = \mathbb{E}[s \mid r]$ is the sentiment expected at rating r , and $\bar{\delta}_i$ is item i 's mean rating-conditioned sentiment residual. With $n_u \geq 5$ and shrinkage $\lambda = 5$, rating leniency and review-tone bias are:

$$\begin{aligned} b_u^R &= \frac{1}{n_u + \lambda} \sum_{i \in \mathcal{D}_u^{\text{tr}}} (r_{ui} - \bar{r}_i^{\text{tr}}), \\ b_u^T &= \frac{1}{n_u + \lambda} \sum_{i \in \mathcal{D}_u^{\text{tr}}} (s_{ui} - \mu_{r_{ui}} - \bar{\delta}_i). \end{aligned} \quad (2)$$

The item mean $\bar{r}_i^{\text{tr}}$ removes the tendency of an item to receive high or low scores. Thus $b_u^R > 0$ indicates a more lenient rating tendency than other users evaluating the same items, whereas $b_u^R < 0$ indicates a stricter rating tendency. Similarly, $b_u^T > 0$ indicates a more positive review tone than expected for the same rating and item, whereas $b_u^T < 0$ indicates a more negative review tone. Both biases are standardized within each dataset, yielding z_u^R and z_u^T .

For interpretation, we set $\tau = 0.5$ and retain the four corner cases in which both standardized biases lie outside $[-\tau, \tau]$: lenient ratings–positive tone, lenient ratings–negative tone, strict ratings–positive tone, and strict ratings–negative tone. Positive and negative tone are defined relative to other reviews with the same rating after item adjustment. For example, a user who rates items more leniently than peers but writes more negatively than typical reviewers assigning the same rating belongs to the lenient-rating–negative-tone case.

We define the user-specific rating–review scale gap (RRS gap) as $D_u = z_u^T - z_u^R$ and its magnitude as $A_u = |D_u|$. Figure 1(a) shows z_u^R and z_u^T on the horizontal and vertical axes, respectively, with the four bias cases and the $D_u = 0$ diagonal. Points above the diagonal have $D_u > 0$, indicating a relatively more positive review-tone tendency than rating tendency, while points below have $D_u < 0$, indicating the opposite. We test whether $\text{sign}(D_u)$ aligns with $\text{sign}(G_{ui})$; D_u represents direction and A_u the magnitude of the difference between the two user scales.

4 Experimental Design

We evaluate the 5-core Amazon Instant Video (IV), Office Products (OP), and Digital Music (DM) datasets [8] using a fixed random 8:1:1 train–validation–test split and ten training seeds (Table 1). For each seed, we compute user-level Spearman correlations between D_u and $\bar{G}_u$, and between A_u and $\bar{M}_u$, both overall and within Q4.

We regress G_{ui} on D_u using ordinary least squares, controlling for log user activity, log item popularity, rating fixed effects, and seed fixed effects; standard errors are clustered by user.

5 Results

Table 2 shows that Neutral outcomes fall from 47.6–68.2% in Q1 to at most 2.4% from Q2 onward. Q4 contains substantial PT and NT in all domains, so both models still win substantial fractions of interactions. Q4 ΔMAE favors NeuMF in IV and OP but ReviewOnly in DM, indicating that average benefit also depends on gain/loss magnitude. Thus, disagreement identifies where selection matters, not which model to trust.

Table 3 shows that training-derived rating and tone biases correlate with held-out validation residuals, suggesting partially persistent user tendencies. Their weak cross-association ($r_s = 0.047$ – 0.102) indicates that numerical leniency and rating-conditioned review tone are non-redundant. We therefore interpret the RRS gap as their standardized relative discrepancy.

Table 3 and Figure 1(b) show that the RRS gap is more strongly associated with prediction direction than prediction-gap magnitude: D_u correlates positively with $\bar{G}_u$ both overall and in Q4, with sign match of 63.3%–82.8% for Q4 interactions with $|D_u| > 0.5$, whereas A_u is only weakly associated with $\bar{M}_u$ ($r_s \leq 0.217$). The positive D_u – G_{ui} association remains after controlling for log user activity, log item popularity, and rating and seed fixed effects, although it is weaker in DM.

Table 4 and Figure 1(c) separate shift direction from benefit. For lenient-rating–negative-tone users, ReviewOnly shifts downward and is less accurate in all domains ($\Delta\text{MAE} +0.110$ to $+0.240$). For strict-rating–positive-tone users, it shifts upward but yields domain-dependent benefit. Thus, the RRS gap characterizes the signed shift, not necessarily an accuracy-improving correction.

6 Discussion and Conclusion

The RRS gap is associated more strongly with the direction than the magnitude of the NeuMF–ReviewOnly shift. Its weaker alignment in DM despite stable biases suggests domain sensitivity, possibly because scalar tone captures less of the review semantics represented by SBERT; this remains to be tested. Because direction does not determine accuracy, the gap is not a standalone selector but could inform a learned gate or an ensemble that weights CF and review predictions. Our simple ReviewOnly expert and three datasets limit generalization; future work will test stronger model pairs, additional domains, and aspect-aware text estimators.

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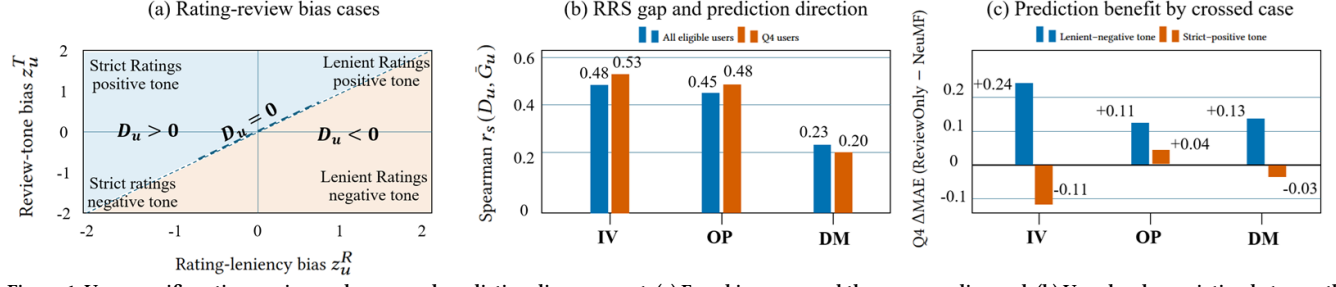


Figure 1: User-specific rating–review scale gaps and prediction disagreement. (a) Four bias cases and the zero-gap diagonal. (b) User-level association between the signed RRS gap D_u and signed prediction gap $\hat{G}_u$. (c) Q4 Δ MAE for the two crossed cases; positive values favor NeuMF and negative values favor ReviewOnly.

Table 1: Statistics of the Amazon 5-core datasets after preprocessing.

Dataset	Users	Items	Interactions	Density (%)
Instant Video	5,130	1,685	37,126	0.429
Office Products	4,905	2,420	53,258	0.449
Digital Music	5,541	3,568	64,706	0.327

Table 2: NeuMF–ReviewOnly error outcomes across prediction-gap quartiles. Δ MAE is ReviewOnly minus NeuMF; positive values favor NeuMF. PT is the percentage of interactions where ReviewOnly absolute error is more than 0.05 lower, NT is the percentage where it is more than 0.05 higher, and Neutral is the remainder.

Dataset	Gap quartile	Δ MAE	ReviewOnly better PT (%)	Similar error Neutral (%)	NeuMF better NT (%)
IV	Q1	+0.001	25.5	47.6	27.0
	Q2	+0.007	47.4	0.8	51.8
	Q3	+0.014	45.7	1.7	52.6
	Q4	+0.035	45.6	2.1	52.3
OP	Q1	+0.002	15.0	68.2	16.8
	Q2	+0.014	44.2	0.6	55.2
	Q3	+0.035	42.0	0.7	57.2
	Q4	+0.062	42.3	1.5	56.2
DM	Q1	+0.002	19.1	60.0	20.9
	Q2	+0.009	46.0	1.0	53.1
	Q3	+0.014	44.8	2.3	52.9
	Q4	−0.028	49.7	2.4	47.9

Table 3: User-bias stability and RRS-gap associations. $D_u = z_u^T - z_u^R$, $A_u = |D_u|$, and $\hat{G}_u$ and $\hat{M}_u$ are user-mean signed prediction gap and magnitude. Seed-varying values are mean $\pm$ SD. Sign match uses Q4 interactions with $|D_u| > 0.5$; β_D and ΔR^2 are from the controlled ordinary least-squares regression.

Dataset	Held-out bias stability		Bias association $r_s(z_u^R, z_u^T)$	Signed-gap direction			Gap magnitude		Controlled regression	
	Rating bias	Tone bias		$r_s(D_u, \hat{G}_u)$ All	$r_s(D_u, \hat{G}_u)$ Q4	Sign match (%)	$r_s(A_u, \hat{M}_u)$ All	$r_s(A_u, \hat{M}_u)$ Q4	β_D	ΔR^2
IV	0.396	0.342	0.068	0.476 ± 0.084	0.529 ± 0.128	82.8 ± 6.7	0.217 ± 0.047	0.091 ± 0.043	0.114	0.181
OP	0.339	0.372	0.102	0.452 ± 0.044	0.483 ± 0.080	82.7 ± 3.6	0.192 ± 0.026	0.149 ± 0.043	0.077	0.143
DM	0.473	0.418	0.047	0.235 ± 0.039	0.200 ± 0.060	63.3 ± 2.6	0.127 ± 0.021	0.126 ± 0.031	0.035	0.021

Table 4: Q4 outcomes for the two crossed rating–review bias cases. Mean G_{ui} is ReviewOnly minus NeuMF; positive Δ MAE favors NeuMF and negative Δ MAE favors ReviewOnly.

Dataset	Lenient ratings, negative tone			Strict ratings, positive tone		
	Mean G_{ui}	Δ MAE	NT (%)	Mean G_{ui}	Δ MAE	PT (%)
IV	−0.540	+0.240	70.8	+0.560	−0.111	55.9
OP	−0.373	+0.110	63.2	+0.397	+0.035	43.7
DM	−0.360	+0.129	65.0	+0.045	−0.029	50.6

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